

Theme 1

Height-sorting machine

Learning Outcomes

By the end of this theme, students will be able to:

- **Identify** key components of a height sorting machine, including sensors, motors, and the sorting mechanism.
- **Explain** the interactions between these components and their interdependence.
- **Recognize** different sensor types (e.g., ultrasonic, infrared) and their applications in sorting machines.
- **Analyse** how coding structures affect the performance and accuracy of the sorting machine..
- **Discuss** the importance of precise measurements and calibration for sorting efficiency.
- **Investigate** the properties of materials used in sorting machines and their suitability for construction.
- **Understand** the relationship between an object's length and sorting criteria.
- **Explore** measurement accuracy principles and their relevance to sorting algorithms.
- **Analyse** factors like sensor placement and motor control that impact sorting effectiveness.
- **Apply** measurement accuracy and motor control principles to optimize the sorting machine.

UNIT 01:

Chapter 01: Lesson 01 : Case Scenario.

Time Frame: 10 MIN

Lesson Objectives:

By the end of the lesson, students will be able to:

Think about the problem of sorting.

Challenge questions of the lesson :

Figure a solution to fix the sorting problem.

Answers of exercises :

- What do you think Adam and Layla might have thought when they saw all those cans piled up?
- Why do you think it's important for a soda factory to sort the cans correctly?
- The problem is that the machine can't sort the cans by *height*. What does it mean to sort things by height?

Suggested Answers :

1. Curiosity, surprise, maybe a little bit of chaos.
2. To make sure they are packaged correctly, to count them easily, to move them to the next steps efficiently.
3. Putting things in order from tallest to shortest, or vice versa.

ESL Relation:

- **Video/Animation:** Provides visual context, making it easier for ESL learners to understand the concept of a height-sorting machine even if they don't understand all the spoken language.
- **Intro Story:** A relatable scenario can capture the interest of ESL learners and provide a shared context for the lesson.
- **Discussion Questions:** The questions are relatively simple and focus on observation and basic understanding, allowing ESL learners to participate even with limited English proficiency. The teacher can facilitate by rephrasing, providing sentence starters, and encouraging peer support.
- **Suggested Answers:** Provide vocabulary and sentence structures that ESL learners can understand and potentially use.

Lesson Plan: Case scenario - The Soda Factory Malfunction (Engage)

Duration: 10 minutes

Objective: To engage students with a real-world problem involving sorting by height and introduce the upcoming design challenge.

1. Engage (10 minutes)

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Teacher Role:

- - Set the stage for the story.
 - Present the case scenario clearly and with enthusiasm.
 - Pose initial thought-provoking questions.
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Expected From Students:

- - Listen attentively to the story.
 - Show curiosity and interest in the problem.
 - Begin to think about potential solutions.
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Intro Story:

- - **Strategy:** Storytelling with visual aid (optional).
 - **Implementation:**
 - Begin by saying something like: "Imagine this: Our friends Adam and Laila had an exciting school trip recently. They went to a real, working soda factory! They got to see all the amazing machines that fill and package the soda cans."
 - Present the case scenario, perhaps displaying a simple image of soda cans piled up: "Everything was going smoothly as they watched the cans moving along the conveyor belts. But then, something unexpected happened! The machine that was supposed to sort the soda cans stopped working correctly. Instead of neatly organized rows, there was a big pile-up of soda cans!"
 - Continue: "The factory manager explained that the current machine isn't able to sort the cans accurately by their height. This is causing big problems – the production is slowed down, and things are getting messy and inefficient."
 - Pause for a moment to let the scenario sink in.

- **Story Reflection:**

- **Questions:**

1. "What do you think Adam and Layla might have thought when they saw all those cans piled up?" (Focus on their observations and potential confusion/interest).
2. "Why do you think it's important for a soda factory to sort the cans correctly?" (Guide towards efficiency, packaging, preventing damage).
3. "The problem is that the machine can't sort the cans by *height*. What does it mean to sort things by height?" (Ensure basic understanding of the concept).

- **Suggested Answers (Guide the discussion towards these points):**

1. Curiosity, surprise, maybe a little bit of chaos.
2. To make sure they are packaged correctly, to count them easily, to move them to the next steps efficiently.
3. Putting things in order from tallest to shortest, or vice versa.

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Strategy: Brief Class Discussion.

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- **Implementation:** Open the floor for a quick minute or two, encouraging students to share their initial reactions and thoughts about the soda factory problem. Acknowledge their ideas and enthusiasm .

UNIT 01:

Chapter 01: Lesson 02 : Engineering Design Process.

Time Frame: 30 MIN

Lesson Objectives:

By the end of the lesson, students will be able to:

Use the Engineering Design Process in designing and building a Height sorting system to fix the problem.

Answers of exercises :

Challenge questions of the lesson :
Figure a solution to fix the sorting problem.

ESL Relation:

- **Visual Aids:** The plan explicitly suggests using a whiteboard/projector and optional visual aids representing the EDP steps. This is crucial for providing non-linguistic support for understanding abstract concepts.
- **Clear and Simple Language:** The Teacher Role emphasizes explaining the steps using simple language, which is essential for ESL learners to grasp new information.
- **Connection to Prior Knowledge:** Starting by reviewing the previous lesson about height sorting provides a familiar context and helps ESL learners connect new vocabulary and concepts to something they already understand.
- **Step-by-Step Approach:** Breaking down the EDP into sequential steps makes the information more manageable for ESL learners.
- **Opportunities for Questions:** Encouraging students to ask clarifying questions is important for ESL learners to address any misunderstandings.

Lesson Plan: Introduction to the Engineering Design Process (Following Height Sorting Challenge)

Duration: 30 minutes

Objective: To introduce students to the basic steps of the engineering design process and how it can be applied to solve the soda can sorting problem.

Materials: Whiteboard or projector, markers, chart paper (optional), visual aids representing the engineering design process steps (optional).

1. Connect & Review (5 minutes)

- **Teacher Role:** Facilitate a brief recap of the previous lesson and introduce the connection to the design process.
- **Expected From Students:** Recall the soda factory problem and their initial thoughts on solutions.
- **Activity:**
 - **Teacher:** "Yesterday, we learned about height sorting and even tried to build our own sorting machines. Remember the story about Adam and Laila at the soda factory? What was the main problem they faced?" (Elicit responses about the malfunctioning height sorting machine).
 - **Teacher:** "Today, we're going to learn a special way that engineers use to solve problems like the one at the soda factory. It's called the Engineering Design Process."
 - **Teacher:** Briefly mention that their task will be to think like engineers to come up with a better solution for sorting those soda cans.

2. Introduce the Engineering Design Process (15 minutes)

- **Teacher Role:** Clearly explain the basic steps of the engineering design process, using simple language and visuals.
- **Expected From Students:** Listen attentively, ask clarifying questions, and begin to relate the steps to the soda can problem.
- **Activity:**
 - **Teacher:** Introduce the key steps of a simplified engineering design process. You can use different variations, but a common one for this age group is:
 1. **Define the Problem:** Clearly state what needs to be solved.
 2. **Brainstorm Ideas:** Come up with as many possible solutions as you can.
 3. **Plan & Select:** Choose the best idea(s) and plan how you would build it.
 4. **Create & Test:** Build a prototype or model of your solution and test if it works.
 5. **Improve & Redesign:** Identify what works and what doesn't, and make changes to improve your design.
 - **Teacher:** Explain each step briefly, relating it back to the soda can sorting problem:
 - **Define the Problem:** "In the soda factory, the problem is that the cans are not being sorted correctly by height, causing delays and inefficiency."
 - **Brainstorm Ideas:** "What are some different ways we could sort the cans by height? Think back to our manual sorting or the video we watched. Don't worry about whether they are perfect ideas yet, just think of possibilities!" (Encourage a few quick examples from students).
 - **Plan & Select:** "Once we have some ideas, we need to choose the one that we think will work best and start planning how we would build it. What materials might we use? How would it work?"

- **Create & Test:** "Then, we would actually build a small version of our idea, a prototype, and test it to see if it sorts the cans correctly."
- **Improve & Redesign:** "After testing, we might find that our first idea wasn't perfect. Engineers always look for ways to make their designs better. We would think about what went wrong and how we could improve our sorting machine."
- **Teacher:** Use visual aids (simple drawings or labels for each step) to reinforce the concepts. You can write the steps on the board or display them on a projector.
- **Teacher:** Encourage students to ask questions about each step.

3. Applying the Design Process to the Soda Can Problem (5 minutes)

- **Teacher Role:** Guide students to start thinking about how the design process applies specifically to the soda can challenge.
- **Expected From Students:** Begin to connect the engineering design process steps to potential solutions for the sorting problem.
- **Activity:**
 - **Teacher:** "Now, let's think about how we can use these steps to solve the soda can problem. What would be the first step we need to take?" (Guide them to "Define the Problem").
 - **Teacher:** "We've already done that! The problem is the malfunctioning height sorting machine."
 - **Teacher:** "What would be the next step?" (Guide them to "Brainstorm Ideas").
 - **Teacher:** "In our next lesson, we will spend more time brainstorming lots of different ideas for how we could build a better height sorting machine for Adam and Laila's soda factory. Think about different ways to move the cans and how you could measure or compare their heights."
 - **Teacher:** Briefly mention that they will then move through the other steps in future lessons.

4. Wrap-up & Preview (5 minutes)

- **Teacher Role:** Summarize the key learning points and prepare students for the next lesson.
- **Expected From Students:** Understand the basic steps of the engineering design process and anticipate the next activity.
- **Activity:**
 - **Teacher:** "So today, we learned about the Engineering Design Process, which is a way that engineers solve problems. We talked about the steps: Define, Brainstorm, Plan, Create, and Improve."
 - **Teacher:** "For our next lesson, your task will be to come ready to brainstorm lots of creative ideas for how we can design and build a new height sorting machine for the soda factory. Think about different materials and ways to move and sort the cans. No idea is a bad idea at this stage!"
 - **Teacher:** Briefly answer any final questions.

Assessment: Observe student participation in discussions and their ability to connect the engineering design process steps to the soda can problem.

Differentiation:

- **For younger students or those needing more support:** Use more visuals and simpler language. Focus on the first two steps (Define and Brainstorm) in more detail.
- **For more advanced students:** Challenge them to think about constraints (e.g., cost, materials available) during the brainstorming phase.

UNIT 01:

Chapter 01: Lesson 03 :Sorting Machine

Time Frame: 90 MIN

Lesson Objectives:

By the end of the lesson, students will be able to:

- Develop the skills needed to measure and cut belts and rolls accurately.
- Demonstrate the ability to use tools and techniques to achieve precise cuts, ensuring proper fit and function.

Challenge questions of the lesson :
Why is sorting soda cans by height important in the factory's production line?

Answers of exercises :

- What is the primary purpose of a height sorting machine? B)
To measure the height of objects and sort them accordingly.
- Which sensor is commonly used with Arduino to measure the height of objects? C)
Ultrasonic sensor.

In an Arduino-based height sorting machine, What does the ultrasonic sensor do? C) It measures the distance to the object, which is used to calculate the height.

ESL Relation:

- **Multi-Sensory Learning:** The session incorporates kinesthetic learning (sorting game), visual learning (video, potential presentation aids), auditory learning (discussions, video audio), and reading (article). This caters to diverse learning styles and strengthens comprehension for ESL learners.
- **Contextualization:** The lesson consistently connects new information to the familiar context of the soda factory problem and the hands-on sorting game.
- **Scaffolding:** The activities progress from a simple game to video observation, reading, assessment, and finally a presentation, providing a structured learning path.
- **Collaboration (Implicit):** While not explicitly a group task in this session, the discussions and potential for group presentations in the "Elaborate" section encourage interaction and language practice.

Lesson Plan: Lesson 3- Exploring Height Sorting and Solutions

Duration: 90 minutes

1. Engage (15 minutes)

- **Teacher Role:**

- Introduce and facilitate the height sorting game.
- Encourage active participation and discussion.
- Connect the game to the previous lesson's challenge.

- **Expected From Students:**

- Participate enthusiastically in the sorting game.
- Communicate their reasoning for placing items in specific bins.
- Make observations about the different heights of objects.

- **Activity - Sorting Challenge:**

- **Strategy:** Hands-on Sorting Game.
- **Implementation:**

1. **Preparation:** Gather a collection of various items as described (blocks, sticks, rods, cylinders, rectangular prisms) and clearly label three bins as "Short," "Medium," and "Tall."
2. **Introduction:** Present the items and bins to the class. Explain the task: "We have these different items, and we have three bins labeled Short, Medium, and Tall. Your challenge is to help me put each item into the bin where you think it belongs based on its height."
3. **Game Play:** Call on individual students or small groups to choose an item and place it in one of the bins. Encourage them to explain their reasoning: "Why did you put that stick in the 'Tall' bin?"
4. **Class Discussion:** After all items are sorted (or during if disagreements arise), facilitate a class discussion:

- **Questions:**

1. "Was it always easy to decide which bin an item belonged in? Why or why not?"
2. "What did you use to help you decide if something was short, medium, or tall compared to the other items?"
3. "Did everyone agree on where each item should go? What does this tell us about sorting by height?"
4. "How is this game similar to the problem Adam and Laila had at the soda factory?"

- **Suggested Answers:**

1. No, sometimes the height differences were small, or the orientation of the object mattered.
2. We compared the items visually.
3. No, there can be some subjectivity. Height is relative.
4. They needed a way to automatically put cans of different heights into the right places.

- **Strategy:** Class Discussion.
- **Implementation:** Open the floor for students to share their experiences and connect the game to the real-world problem.

2. Explore (20 minutes)

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Teacher Role:

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- Introduce the video and article.
- Set a purpose for watching and reading.
- Facilitate brief discussions after each.

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Expected From Students:

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- Watch the video actively and identify key components and processes.
- Read the article attentively and extract information about height sorting.
- Share their observations and understanding.

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Activity - Watch It:

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- **Strategy:** Video Observation and Discussion.
- **Implementation:**

2. **Pre-Watching:** "Now that we've tried sorting by height ourselves, let's watch a short video to see how real height sorting machines work in factories. Pay attention to how the machine measures the height and how it moves the objects into different groups."
3. **During Watching:** Play a short, engaging video showcasing a height sorting machine in action (e.g., sorting packages, products on a conveyor belt).
4. **Post-Watching:**

- **Questions:**

1. "What did you notice about how the machine measured the height of the objects?"
2. "How did the machine move the objects into different height categories?"
3. "Were there any parts of the machine that looked familiar to ideas we might have for our own sorting machine?"
4. "What were some of the benefits of using a machine to sort by height that you saw in the video?"

- **Strategy:** Think-Pair-Share (optional) followed by whole-class discussion.

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Activity - Read About It:

- - **Strategy:** Guided Reading and Information Extraction.
 - **Implementation:**
 2. **Pre-Reading:** "To understand more about why height sorting is important and how it's used in different industries, let's read a short article together." (Provide a concise and age-appropriate article about the applications of height sorting in various fields like manufacturing, logistics, or agriculture).
 3. **During Reading:** Read the article aloud as a class or have students read in pairs. Encourage them to highlight or note down key information about the benefits and applications of height sorting.
 4. **Post-Reading:**
 - **Questions:**
 1. "According to the article, where might you find machines that sort objects by height?"
 2. "What are some reasons why sorting by height is useful in these places?"
 3. "Did the article mention any challenges or important considerations when designing a height sorting system?"
 4. "How does the information in the article connect to what we saw in the video?"

3. Explain (15 minutes)

- **Teacher Role:**
 - Clarify key concepts related to height measurement and sorting mechanisms.
 - Address any misconceptions arising from the game, video, or article.
 - Introduce the idea of sensors and their role in automated height sorting.
- **Expected From Students:**
 - Articulate their understanding of height sorting principles.
 - Ask clarifying questions.
 - Begin to connect the observed mechanisms to potential design solutions.
- **Concept Clarification:**
 - **Teacher:** "From our game and the video, we saw that sorting by height involves comparing how tall things are. But how can a machine do that accurately and quickly?"
 - **Teacher:** Briefly explain different ways height can be measured automatically (e.g., using light beams, ultrasonic sensors – connect back to the assessment questions if appropriate, without giving away answers yet).
 - **Teacher:** Discuss different mechanisms for moving sorted items (e.g., conveyor belts with pushers, ramps with different heights, gates that open and close).
 - **Teacher:** "Think about the soda factory again. Why would it be important for their sorting machine to be accurate and fast?" (Efficiency, preventing errors in packaging).

4. Evaluate (20 minutes)

- **Teacher Role:**

- Administer the short assessment.
- Ensure students understand the questions.
- Collect and review responses to gauge understanding.

- **Expected From Students:**

- Answer the assessment questions thoughtfully based on what they have learned.
- Demonstrate their understanding of the purpose and mechanisms of height sorting.

- **Activity - Choose the Answers:**

- **Strategy:** Short Multiple-Choice Assessment.
- **Implementation:** Provide students with the following questions (or similar ones):
 2. **What is the main goal of a machine that sorts objects by height?** A) To weigh objects. B) To measure the height of objects and separate them into different groups based on their height. C) To change the shape of objects. D) To count the number of objects.
 3. **Which of these might be used by a machine to "see" or measure the height of an object without touching it?** A) A scale. B) A camera and computer vision. C) A heater. D) A container.
 4. **Imagine a height sorting machine with three exits: one for short items, one for medium items, and one for tall items. What would happen to a medium-sized object?** A) It would go to the short exit. B) It would go to the tall exit. C) It would go to the medium exit. D) It would get stuck.
- **Teacher:** Allow students a few minutes to answer the questions individually. Review answers briefly as a class after collection, addressing any common misunderstandings.

5. Elaborate (20 minutes)

- **Teacher Role:**

- Introduce the presentation task.
- Guide students to think about the real-world impact of height sorting.
- Provide examples and structure for their presentations.

- **Expected From Students:**

- Work individually or in small groups to prepare a short presentation.
- Research or brainstorm examples of height sorting in industry.
- Explain the benefits and importance of height sorting in those contexts.

- **Activity - Make a Presentation:**

- **Strategy:** Short Presentation Preparation.
- **Implementation:**
 1. **Introduction:** "Now that we understand how height sorting works and why it's important, let's think about how it's actually used in big industries. Your task is to prepare a very short presentation (1-2 minutes per person/group) about how height sorting helps in different industries."
 2. **Guidance:**

- **Teacher:** "Think about the article we read and the video we watched. Where might height sorting be useful?" (Prompt with examples like factories, warehouses, food processing, recycling).
- **Teacher:** "For your presentation, you can choose one industry and explain:"
 - What kind of objects are being sorted by height?
 - Why is it important to sort them by height in this industry?
 - What are the benefits of using height sorting machines? (e.g., efficiency, quality control, organization).
- **Teacher:** "You can create a simple poster, draw a picture, or just speak to the class. Focus on clearly explaining the 'why' and 'how' of height sorting in your chosen industry."

3. **Preparation Time:** Allow the remaining time for students to briefly brainstorm and prepare their presentations. Inform them that they will share these in the next lesson.

Conclusion & Homework Assignment (5 minutes)

- **Teacher Role:**

- Summarize the key learning points of the session.
- Assign homework related to the presentation preparation.

- **Expected From Students:**

- Reflect on what they have learned about height sorting.
- Continue working on their presentation.

- **Homework/Extension:**

- **Teacher:** "For homework, please continue to think about your chosen industry and gather any more information or make notes for your short presentation on how height sorting is helpful there."

- **Assessment:**

- **Formative:** Observation of student participation in the game and discussions, review of assessment answers.
- **Assumptive:** Preparation for the presentations in the next lesson will further reveal their understanding.

Relation to Standards:

- NGSS (MS-ETS1-1, MS-ETS1-2, MS-ETS1-3)
- CCSS (6.SP.B.5, 7.SP.A.1)
- ISTE (6b, 7b)

UNIT 01:

Chapter 01: Lesson 04 : Design and Build Height Sorting Machine.

Time Frame: 90 MIN

Lesson Objectives:

By the end of the lesson, students will be able to:

- Understand the functionality of a height-based sorting machine.
- Design and create a basic prototype that sorts objects based on their height.
- Apply the engineering design process to prototype and test a sorting machine.
- Develop teamwork and problem-solving skills while creating the prototype.

Answers of exercises :

Suggested Answers:

- Height involves linear measurement, while color and weight involve different properties.
- Warehouses, recycling plants, manufacturing.
- Efficiency, organization, quality control.

Challenge questions of the lesson :

- What do you think makes height-based sorting different from other types of sorting (like by color or weight)?"
- "Where might we see machines that sort objects by height?"
- "Why do you think height might be an important characteristic to sort objects by?"

ESL Relation:

- **Visual Support:** Use videos, diagrams, and real objects to aid comprehension.
- **Simplified Language:** Rephrase complex questions and instructions.
- **Hands-On Activities:** Emphasize practical tasks to minimize language reliance.
- **Collaborative Learning:** Encourage group work for peer support and language practice.
- **Modeling and Demonstrations:** Show students how to perform tasks.
- **Graphic Organizers and Sentence Frames:** Provide visual and linguistic support.
- **Vocabulary Pre-teaching and Visual Glossaries:** Introduce and reinforce key terms.
- **Chunking and Think-Alouds:** Break down complex information and model thinking.
- **Alternative Assessments:** Allow for visual presentations and demonstrations.
- **Clear, Written Instructions:** Provide accessible homework directions.

Lesson Plan : Lesson 4 : Design and Build Height Sorting Machine.

Duration: 90 minutes

1. Engage (10 minutes)

- **Teacher Role:**

- Facilitate class discussions.
- Pose thought-provoking questions.
- Introduce the lesson's goal.

- **Expected From Students:**

- Participate in discussions.
- Share ideas and observations.
- Ask questions.

- **Intro Story:**

- Strategy: Video/Animation Presentation.

1. Implementation: Show a short video of a height-sorting machine in a real-world setting (warehouse, factory).

- **Story Reflection:**

- Questions:

1. "What do you think makes height-based sorting different from other types of sorting (like by color or weight)?"
2. "Where might we see machines that sort objects by height?"
3. "Why do you think height might be an important characteristic to sort objects by?"

- Suggested Answers:

1. Height involves linear measurement, while color and weight involve different properties.
2. Warehouses, recycling plants, manufacturing.
3. Efficiency, organization, quality control.

- Strategy: Class Discussion.

1. Implementation: Open the floor for students to share their thoughts and build upon each others responses.

2. Explore (20 minutes)

- **Teacher Role:**

- Guide hands-on exploration.
- Facilitate group discussions.
- Provide materials and support.

- **Expected From Students:**

- Manually sort objects by height.
- Brainstorm sorting methods.

- Sketch design ideas.
- **Activity - Do It:**
 - Materials Needed: Various objects of different heights, rulers.
 - Step-by-Step Instructions:
 1. Present students with various objects.
 2. Instruct students to manually sort objects by height using rulers.
 3. Facilitate a discussion about the challenges of manual sorting.
 - Reflection Questions:
 - What challenges did you face when manually sorting the objects?
 - What are some different ways we could sort the objects?
- **Brainstorm Web:**
 - Space for ideas and connections related to height sorting methods (ramps, platforms, conveyors).

3. Explain (20 minutes)

- **Teacher Role:**
 - Explain the concept of height sorting.
 - Introduce key components and design principles.
 - Show a diagram of a basic height sorting system.
- **Expected From Students:**
 - Engage actively in discussions.
 - Ask clarifying questions.
 - Take notes.
- **Watch It:**
 - Pre-Watching: Set a purpose by introducing the concepts of conveyor belts and ramps.
 - During Watching: Active engagement by having students take notes on the function of each component.
 - Post-Watching: Discussion and reflection on how the video relates to the lesson's objective.
- **Read About It:**
 - Pre-Reading: Discussion on real world applications of height sorting.
 - During Reading: Engagement by having students highlight key concepts.
 - Post-Reading: Summarize and reflect on the design principles of a height sorting system.

4. Elaborate (25 minutes)

- **Teacher Role:**

- Encourage creative design solutions.
- Provide support and guidance during construction.
- Ask probing questions to stimulate critical thinking.

- **Expected From Students:**

- Design and build prototype height-sorting machines.
- Experiment with different materials and mechanisms.
- Work collaboratively in groups.

- **Practice:**

- Materials Needed: Various construction materials (cardboard, tape, etc.), objects of varying heights.
- Step-by-Step Instructions:

1. Groups design and build their prototypes.
2. Students test and refine their designs.
3. Teacher provides feedback and support.

- Reflection Questions:

- What challenges did you encounter during construction?
- How did you solve those challenges?
- What improvements could you make?

- **Challenge:**

- Homework/Extension: Research real-world sorting systems and write a report.

5. Evaluate (15 minutes)

- **Teacher Role:**

- Facilitate prototype testing and demonstrations.
- Lead class discussions on prototype effectiveness.
- Assess student understanding and design choices.

- **Expected From Students:**

- Demonstrate prototype functionality.
- Explain design choices and reasoning.
- Participate in class discussions.

- **Showcase:**

- Evaluation: Assess prototype functionality, design creativity, and explanation clarity.
- Rubric:

- Functionality (Accuracy, Efficiency): (4 points)

- Design Creativity (Innovation, Resourcefulness): (3 points)
- Explanation (Clarity, Understanding): (3 points)

6. Conclusion & Homework Assignment (10 minutes)

- Teacher Role:
 - Summarize key learning points.
 - Assign homework.
- Expected From Students:
 - Complete exit ticket.
 - Begin homework assignment.
- Homework/Extension:
 - Research real-world sorting systems and write a brief report.
- Assessment:
 - Formative: Observation of group participation and design process.
 - Assumptive: Evaluation of prototype functionality and student explanations.

🚗 Relation to Standards:

- NGSS (MS-ETS1-1, MS-ETS1-2, MS-ETS1-3)
- CCSS (6.SP.B.5, 7.SP.A.1)
- ISTE (6b, 7b)

UNIT 01:

Chapter 01: Lesson 05 :Mechanics of sorting machine

Time Frame: 90 MIN

Lesson Objectives:

By the end of the lesson, students will be able to:

- Familiar with complex machines (gears systems) and their principles.
- Choose the best material and design the machine.
- Choose the best type, material and design for the machine.
- Utilize Fusion 360 to create detailed 3D models of the height sorting machine components, focusing on precision and functionality to enhance the design process.

Challenge questions of the lesson :

- How do different gears in a gearbox work together to change the speed and torque of a machine?
 - What are some real-life examples where complex gearboxes are used, and why are they important in those applications ?
 - How can you model a simple gearbox using software design applications?
 - What does changing the gear ratio in a gearbox affect?
 - If you have a gearbox with a large gear driving a small gear, What happens to the speed and torque?
3. Why might a complex gearbox use multiple gears?
4. If a driving gear has 10 teeth and the driven gear has 20 teeth, what is the gear ratio (driven:driving)?

Answers of exercises :

- Different gears in a gearbox change speed and torque through their tooth ratio, where smaller driving larger driven reduces speed (increases torque) and vice versa.
- Complex gearboxes are used in vehicles, wind turbines, and industrial machinery to match the power source's output to the application's needs for varying speed and torque.
- A simple gearbox can be modeled in software design applications by representing gears as circles or 3D models and simulating their meshing and rotation.

Answers to assesment questions:

- B) The speed and torque of the machine.
- B) The speed increases, and the torque decreases.
- B) To provide different speed and torque settings for various tasks.
- 4B) 2:1.

ESL Relation:

- **Multi-Sensory Learning:** The lesson incorporates visual learning (Fusion 360, video, interactive tool), kinesthetic learning (manipulating gears in the interactive tool, future prototype construction), and auditory learning (discussions, video audio).
- **Contextualization:** The lesson builds directly on the previous lessons about height sorting and introduces gears as a potential solution, maintaining a relevant context.
- **Scaffolding:** The activities progress from guided digital design to video observation, interactive exploration, assessment, and finally a more complex design task, providing a structured learning path.
- **Active Learning:** The hands-on and interactive nature of the activities promotes active engagement, which is beneficial for language acquisition as it provides opportunities for practical application.

Lesson Plan: Lesson 5 - Designing and Exploring Gear Systems

Duration: 90 minutes

1. Engage (15 minutes)

- **Teacher Role:**
 - Introduce the gear design activity using Fusion 360.
 - Guide students through the basic gear creation process.
 - Encourage experimentation with different gear sizes.
- **Expected From Students:**
 - Open Fusion 360 and follow instructions to create basic gears.
 - Experiment with changing the number of teeth and observe the visual differences in gear size.
 - Ask questions about the design process.
- **Activity - Digital Gear Design:**
 - **Strategy:** Guided Software Exploration.
 - **Implementation:**
 1. **Introduction:** "Yesterday, we learned how gears work to change speed and torque. Today, we're going to start designing our own gears using Fusion 360. This will help us understand how the size of a gear affects its function."
 2. **Step-by-Step Guidance:** Provide clear, step-by-step instructions on how to use the gear creation tool within Fusion 360 (or a similar CAD software). Focus on parameters like the number of teeth and how changing this value alters the gear's diameter.
 3. **Experimentation:** Encourage students to create several gears with varying numbers of teeth (e.g., a gear with 20 teeth and another with 40 teeth).
 4. **Observation:** Ask students to visually compare the sizes of the gears they have designed. "What do you notice about the size of the gear when you increase the number of teeth?"
 - **Strategy:** Think-Pair-Share.
 - **Implementation:** Have students discuss their observations with a partner before sharing with the whole class.

2. Explore (20 minutes)

- **Teacher Role:**
 - Introduce the video about gear systems.
 - Guide students to focus on how gears interact and the effects of different gear ratios.
 - Facilitate the interactive gear angle manipulation activity.
- **Expected From Students:**
 - Watch the video attentively and identify different types of gear arrangements and their functions.
 - Actively manipulate the gear angles in the interactive tool and observe the resulting changes in movement.
 - Discuss their observations and relate them to speed and torque.
- **Activity - Watch It:**

- **Strategy:** Video Observation and Discussion.
- **Implementation:**

1. **Pre-Watching:** "Now, let's watch a video that shows different types of gears working together in gearboxes. Pay attention to how the sizes of the gears affect the speed and direction of rotation."
2. **During Watching:** Play a short, informative video showcasing various gear types (spur, bevel, worm, rack and pinion) and how they are used to change speed and torque.
3. **Post-Watching:**

- **Questions:**

1. "What were some of the different types of gears you saw in the video?"
2. "How did the video demonstrate the relationship between the size of the driving and driven gears and their speeds?"
3. "Did you see any examples of how gears can change the direction of motion?"

- **Activity - Interactive Gear Manipulation:**

- **Strategy:** Digital Simulation/Interactive Tool.
- **Implementation:**

1. **Introduction:** "Now, we're going to use a simple interactive tool to explore what happens when we change the angles of gears that are connected." (Provide access to a simple online gear simulator or a pre-prepared interactive model).
2. **Guided Exploration:** Instruct students on how to manipulate the angles of the gears in the tool.
3. **Observation and Discussion:**

- **Teacher:** "Try rotating one of the gears. What happens to the gear it's connected to?"
- **Teacher:** "Now, try changing the angle at which the gears are connected. What do you observe about their ability to move together smoothly?"
- **Questions:**

4. "What happens if the teeth of the gears are not properly aligned?"
5. "Why is the angle at which gears are mounted important for a gearbox to work correctly?"
6. "How does this relate to the design of a real gearbox?"

- **Strategy:** Small Group Discussion.
- **Implementation:** Have students discuss their observations in small groups and formulate explanations.

3. Explain (15 minutes)

- **Teacher Role:**

- Reinforce the concepts of gear ratio, speed, and torque.
- Explain the importance of proper gear meshing and alignment.
- Introduce the concept of rack and pinion gears in preparation for the practice activity.

- **Expected From Students:**

- Articulate their understanding of the relationship between gear size, speed, and torque.
- Explain why gear alignment is crucial.
- Understand the basic function of a rack and pinion system.

- **Concept Reinforcement:**

- **Teacher:** "Let's revisit what we learned about gear ratios. Remember, the ratio of the number of teeth on the driven gear to the driving gear tells us how the speed and torque will change."
- **Teacher:** Use examples from the video and the Fusion 360 designs to illustrate different gear ratios and their effects.
- **Teacher:** "From our interactive tool, we saw that the angle at which gears are connected is very important. If the teeth don't mesh properly, the gears won't turn smoothly, and the system won't work efficiently or could even break."
- **Teacher:** Introduce rack and pinion gears: "Another important type of gear system is the rack and pinion. A rack is a straight, toothed bar, and a pinion is a circular gear. When the pinion rotates, the rack moves linearly, or vice versa. Can you think of any real-world examples where we might see a rack and pinion system?" (Steering systems in cars, some types of linear actuators).

4. Evaluate (20 minutes)

- **Teacher Role:**

- Administer the short multiple-choice assessment.
- Ensure students understand the questions and answer choices.
- Review answers to gauge understanding of gear principles.

- **Expected From Students:**

- Answer the assessment questions based on their understanding of gear ratios, speed, torque, and gear system design.

- **Activity - Choose the Correct Answer:**

- **Strategy:** Short Multiple-Choice Assessment.
- **Implementation:** Provide students with the following questions:
 1. **What does changing the gear ratio in a gearbox affect?** A) The color of the machine. B) The speed and torque of the machine. C) The temperature of the machine. D) The shape of the machine.
 2. **If you have a gearbox with a large gear driving a small gear, what happens to the speed and torque?** A) The speed decreases, and the torque increases. B) The speed increases, and the torque decreases. C) The speed and torque both stay the same. D) The speed decreases, and the torque also decreases.
 3. **Why might a complex gearbox use multiple gears?** A) To make the machine look more interesting. B) To provide different speed and torque settings for various tasks. C) To increase the noise of the machine. D) To reduce the machine's efficiency.
 4. **If a driving gear has 10 teeth and the driven gear has 20 teeth, what is the gear ratio (driven:driving)?** A) 1:2. B) 2:1. C) 1:1. D) 2:3.
- **Teacher:** Allow students time to answer individually and then review the correct answers as a class, explaining the reasoning behind each.

5. Elaborate (15 minutes)

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- **Teacher Role:**

- Introduce the rack and pinion design practice in Fusion 360.
- Provide guidance and support as students create their designs.
- Connect this activity to potential applications in their sorting machine.

- **Expected From Students:**

- Apply their knowledge of gear design to create a rack and pinion system in Fusion 360.
- Experiment with the movement of the rack and pinion.
- Begin to think about how gears could be used in their sorting machine design.

- **Activity - Practice: Rack and Pinion Design:**

- **Strategy:** Guided Software Practice.
- **Implementation:**
 1. **Instruction:** "Now that we know about rack and pinion gears, let's try designing one in Fusion 360. This type of gear system can be very useful for creating linear motion from rotational motion."
 2. **Step-by-Step Guidance:** Provide instructions on how to create a rack and a pinion gear and then assemble them so that their teeth mesh correctly.
 3. **Experimentation:** Encourage students to try rotating the pinion and observe the linear movement of the rack.
 4. **Connection to Sorting Machine:** "Think about how a rack and pinion system could potentially be used in your height sorting machine. Could it help move objects or activate a sorting mechanism?"

Conclusion & Homework Assignment (5 minutes)

- **Teacher Role:**

- Summarize the key concepts covered in the lesson.
- Introduce the final design challenge for the following sessions.

- **Expected From Students:**

- Reflect on their learning about gear design and function.
- Understand the upcoming design and construction task.

- **Homework/Extension:**

- **Teacher:** "Today, we explored the design and function of different gears. For the next few sessions, your main task will be to **design and construct a prototype sorting mechanism using simple machines like levers and pulleys**. You should also **select suitable materials for building your sorting machine, considering their mechanical properties**. Think about how you might incorporate the principles of **gear systems** that we've learned about to help with the sorting process. Finally, consider the **importance of stability** in any moving parts or the overall structure of your machine."

- **Assessment:**

- **Formative:** Observation of student engagement in Fusion 360 and discussions, review of assessment answers.
- **Assumptive (Ongoing):** The final sorting machine prototype will be the primary assumptive assessment.

Relation to Standards:

- NGSS (MS-ETS1-1, MS-ETS1-2, MS-ETS1-3)
- CCSS (6.SP.B.5, 7.SP.A.1)
- ISTE (6b, 7

UNIT 02:

Chapter 01: Lesson 01 :Add the sensor

Time Frame: 90 MIN

Lesson Objectives:

- 1-Understand the fundamental principles of infrared (IR) and ultrasonic sensor technologies for object detection and distance measurement.
- 2-Identify potential advantages and limitations of both IR and ultrasonic sensors for height measurement.
- 3-Build a basic circuit incorporating an IR sensor, LED, and power source on a breadboard.
- 4-Understand the importance of correct component polarity in circuits.
- 5-Begin to apply the Engineering Design Process (EDP) by considering the integration of an IR sensor into a height sorting machine design.

Challenge questions of the lesson :

- 1-In what situations might an IR sensor be a better choice for distance measurement than an ultrasonic sensor, and vice versa?
- 2-What are some real-world factors (like object material, surface, or environment) that could affect the accuracy of IR or ultrasonic sensors in a height sorting machine?
- 3-Why is it important to pay close attention to the polarity of electronic components when building circuits?
- 4-How does understanding the characteristics of a sensor help in the "create" stage of the Engineering Design Process for our height sorting machine?

Here are short answers to the challenge questions:

- **IR better than ultrasonic:** For detecting very small objects or when silent operation is needed.
- **Ultrasonic better than IR:** For more accurate distance in various lighting conditions and with transparent/dark objects.
- **Factors affecting accuracy:** Object material (absorption/reflection), surface (smooth/rough), and environment (ambient light for IR, temperature/humidity for ultrasonic).
- **Importance of polarity:** Incorrect polarity can prevent components from working or even damage them.
- **Understanding sensors helps in "create":** It informs where to place the sensor, how to orient it, and how to interpret its data for accurate sorting

SEL Relation:

- **Self-Awareness:** Students reflect on their understanding of sensor technologies and their ability to follow instructions for circuit building.
- **Self-Management:** Students practice carefulness and precision while building circuits and recording observations.
- **Social Awareness:** Students may collaborate during the "Explore" and "Practice" activities, sharing observations and troubleshooting circuit issues.
- **Relationship Skills:** Students communicate their findings and ideas during class discussions.
- **Responsible Decision-Making:** Students learn about the importance of correct procedures in electronics (circuit building) and informed choices in technology selection (sensor type).

1. Engage:

Teacher Role:

- Start a discussion about how machines automatically sort things in the real world (like packages or recycling), linking it to what students might already know.
- Show interesting pictures or videos of these automated sorting systems to grab their attention.
- Ask students to think about what problems you might face and what good things could come from using machines to sort objects by height.
- Lead a discussion on why measuring height correctly is really important for sorting to work well.
- Help students connect the idea of measuring height with things they already know and then introduce the idea of using sensors to do this automatically and precisely.

Expected From Students:

- Share examples of automatic sorting they've seen or can imagine.
- Watch the pictures or videos of sorting machines and share their first thoughts and questions about how they work.
- Think about possible issues and benefits of using machines to sort by height.
- Discuss why getting the height measurement right is key for sorting.
- Connect what they already know about measuring height to the idea of using sensors in a machine.

Strategies to use with this part :

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Preview with a Hook:

- o Strategy: Start with a surprising question or a cool picture about automatic sorting.
- o Implementation:
 - Show a short video of packages moving super fast on belts and being sorted into different bins.

- Ask, “How do these machines know where each package needs to go? What kind of information do you think the machine uses to decide?”
 - Briefly talk about it, guiding them to think about things like size.
- **Make Predictions:**
 - o Strategy: Tell them what the lesson is about and the first thing they'll learn regarding IR sensors.
 - o Implementation:
 - Say the lesson is called “Add the sensor.”
 - Read the first thing they'll learn: "Understand how infrared (IR) technology works and how it's used to see objects."
 - Ask students, “Based on the lesson name and what you'll learn, what do you think we'll be adding to our height sorting machine? What do you already know or can guess about infrared (IR)?” Write down a few of their ideas.
- **Set the Scene with a Visual:**
 - o Strategy: Use the pictures of different sensors from the page.
 - o Implementation:
 - Show the six sensor pictures.
 - Ask students, “We need a way for our machine to measure how tall something is without touching it. Look at these different sensors. Which ones do you think could ‘see’ how tall an object is from a distance?” Ask them to say why they picked those ones based on what they look like.
- **Think-Pair-Share:**
 - o Strategy: Have students talk to each other about the problems of sorting by height.
 - o Implementation:
 - Ask, “Imagine you're designing a machine to quickly and correctly sort objects of different heights. What are some of the tricky things you think you'd have to deal with?”
 - Have them talk about it with the person next to them.
 - Ask some students to share their ideas with the whole class.

2-Explore

Teacher Role:

- Prior Preparation: Before the lesson, prepare Arduino boards with the provided code (or a similar code that reads both IR and ultrasonic sensors and outputs the raw/calculated values to the serial monitor). Ensure each setup has the necessary jumper wires readily available.
- Guide students to correctly connect the provided IR sensor and ultrasonic sensor to the pre-loaded Arduino boards using breadboards and jumper wires, following clear instructions or diagrams.
- Ensure students understand how to open the Arduino Serial Monitor to view the sensor readings.
- Encourage careful observation of how the sensor readings change as objects are placed at different heights or distances from each sensor.
- Facilitate a discussion comparing the behavior and potential applications of the two sensor types based on their observations.

Expected From Students:

- Correctly connect the IR sensor and ultrasonic sensor to the pre-loaded Arduino board according to instructions.
- Open and monitor the Arduino Serial Monitor to observe the data being output by each sensor.
- Experiment with placing objects at varying distances from both the IR and ultrasonic sensors.
- Observe and note down the differences in how the readings from each sensor change in response to the object's presence and distance/height.
- Participate in a discussion comparing the characteristics of the IR and ultrasonic sensors for height measurement.

Materials Needed:

- Student books with this "Explore" page.
- Arduino microcontrollers (pre-loaded with appropriate code).
- Breadboards and jumper wires.

- IR sensors suitable for distance measurement (with connection diagrams/instructions).
- Ultrasonic sensors (e.g., HC-SR04) (with connection diagrams/instructions).
- Objects of varying heights to test with the sensors.
- Student notebooks or a designated space for recording observations.
- Access to computers with the Arduino IDE installed (primarily for viewing the Serial Monitor).

Arduino code :

```
// Define sensor pins
const int irSensorPin = A0;    // Example analog pin for IR sensor
const int trigPin = 9;        // Trigger pin for ultrasonic sensor
const int echoPin = 10;       // Echo pin for ultrasonic sensor

void setup() {
  Serial.begin(9600);
  Serial.println("IR and Ultrasonic Sensor Distance Readings");
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
}

void loop() {
  // Read IR sensor value
  int irValue = analogRead(irSensorPin);
  Serial.print("IR Value: ");
  Serial.println(irValue);
  Serial.print("Approximate IR Reading (Qualitative - depends on sensor):");
  // The relationship between analog value and distance varies greatly
  // depending on the specific IR sensor. You'll need to experiment
  // and potentially calibrate to get meaningful distance in cm or inches.
  // For now, we'll just show the raw analog reading.
```

```

Serial.println(irValue);

// Measure distance with ultrasonic sensor
long duration = measureDistance();
// Approximation for distance in centimeters (adjust based on speed of
sound and your sensor)
int distanceCm = duration * 0.034 / 2;
Serial.print("Ultrasonic Distance (cm): ");
Serial.println(distanceCm);

Serial.println("-----");
delay(500); // Delay between readings
}

long measureDistance() {
// Clear the trigPin by setting it LOW for a short duration
digitalWrite(trigPin, LOW);
delayMicroseconds(2);

// Trigger the sensor by setting the trigPin HIGH for 10 microseconds
digitalWrite(trigPin, HIGH);
delayMicroseconds(10);
digitalWrite(trigPin, LOW);

// Measure the duration of the echo pulse
return pulseIn(echoPin, HIGH);
}

```

Step-by-Step Instructions:

1. Introduction to the Exploration (10 minutes):
 - Explain that the Arduinos have already been programmed to read data from both an IR and an ultrasonic sensor.
 - Their task is now to connect each sensor to the Arduino and observe

how they work.

2. Circuit Connection (20-30 minutes):

- Provide clear, step-by-step instructions and/or diagrams for connecting the IR sensor and the ultrasonic sensor to the Arduino. Emphasize the importance of correct wiring to the specified pins.
- Guide students through the wiring process, ensuring they understand each connection.

3. Observing Sensor Readings (20-30 minutes):

- Instruct students on how to open the Arduino IDE and then open the Serial Monitor (Tools -> Serial Monitor). Ensure the baud rate is set correctly (usually 9600).
- Explain that they will see numerical data appearing in the Serial Monitor, representing the readings from each sensor.
- Guide them to experiment by placing different objects at various heights or distances from each sensor and carefully observing how the "IR Value" and "Ultrasonic Distance (cm)" readings change.
- Encourage them to note down their observations for each sensor:
 - How the numbers change as the object gets closer or farther away.
 - The range over which each sensor seems to detect objects.
 - Any noticeable differences in the behavior of the two sensors.

4. Discussion and Comparison (15 minutes):

- Facilitate a class discussion based on their observations. Prompts could include:
 - "What differences did you notice in how the IR sensor and the ultrasonic sensor reacted to the objects?"
 - "Which sensor seemed to give a more consistent reading for distance?"
 - "Were there any situations where one sensor seemed to work better than the other?"

- "Based on what you observed, which type of sensor do you think might be more suitable for accurately measuring the height of objects for our sorting machine, and why?"

Differentiation:

- Support: Provide students who struggle with circuit connections with individual assistance and ensure they are following the diagrams correctly.
 - Extension: Challenge students who quickly understand the sensor behavior to think about potential limitations of each sensor in a real-world sorting machine scenario (e.g., influence of object material or color on IR, sensitivity to soft materials for ultrasonic).
-

3-Assessment

Teacher Role:

- Introduce the "Focus" section as a way to check students' understanding of the characteristics and limitations of IR and ultrasonic sensors.
- Read each question aloud clearly to the students.
- Ensure students understand the multiple-choice format and how to select their answers (circling the letter).
- Encourage students to answer based on their observations from the "Explore" activity and the information provided.
- Review the correct answers as a class after students have completed this section, explaining the reasoning behind each.

Expected From Students:

- Listen attentively as the teacher reads each question.
- Recall their experiences from the "Explore" activity and the information they learned about IR and ultrasonic sensors.
- Critically evaluate the provided options and select the most accurate answer for each question.

- Circle the letter corresponding to their chosen answer.
- Participate in the review of the answers and listen to the teacher's explanations.

Materials Needed:

- Student books with this "Assessment" page.
- Pencils for circling answers.

Step-by-Step Instructions:

1. Introduction to "Focus" (5 minutes):

- Say something like: "Now let's check what we've learned about the two types of sensors we've been working with. Read each question carefully and choose the best answer based on what you observed and what you've learned."

2. Administering the Questions (10-15 minutes):

- Read question 1: "Which sensor type is generally more accurate for precise distance measurements?"
- A) IR sensor
- B) Ultrasonic sensor
- C) Both are equally accurate
- D) Neither is accurate for distance measurements.
- (Correct Answer: B)
- *Teacher Note: Explain that while both have their uses, ultrasonic sensors are often considered more accurate for precise distance measurements in many applications due to their method of measuring time-of-flight of sound waves.*

- Read question 2: "What is a potential limitation of IR sensors?"
- A) They can't work in dark environments
- B) They are affected by ambient light
- C) They can't detect objects at all
- D) They only work with metal objects.
- (Correct Answer: B)
- *Teacher Note: Discuss how strong sunlight or other IR sources can interfere with the readings of IR distance sensors.*
-
- Read question 3: "Ultrasonic sensors measure distance by:"
- A) Analyzing color patterns
- B) Detecting heat signatures
- C) Measuring the time it takes for sound waves to bounce back
- D) Counting the number of light reflections.
- (Correct Answer: C) –
- *Teacher Note: Reinforce the principle of ultrasonic sensors emitting sound waves and measuring the echo return time*
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3. Student Response (5-10 minutes):

- Instruct students to circle the letter of the answer they think is correct for each question. Allow sufficient time for them to consider their choices.

4. Review and Discussion (10 minutes):

- Once everyone has finished, go through each question and reveal the correct answer.
- For each question, briefly explain *why* the correct answer is right and why the other options are not the best fit, referencing their experiences from the "Explore" activity if applicable.

Differentiation:

- **Support:** For students who are struggling, you can prompt them by asking questions related to their observations during the "Explore" activity. For example, "Did the IR sensor readings seem to change if you moved a bright light near it?"
 - **Extension:** Challenge students to research real-world applications where either IR or ultrasonic sensors are preferred due to their specific characteristics.
-

4-Practice

Teacher Role:

- Guide students to build a simple circuit using an IR sensor, LED, resistor, battery, and wires on a breadboard. Provide clear circuit diagrams and instructions.
- Emphasize the importance of correct polarity for the IR sensor and the LED.
- Instruct students to carefully observe what happens when the IR sensor's terminals are reversed.
- Transition the lesson back to the Engineering Design Process, guiding students to consider how they will integrate the IR sensor into the "create" stage of their height sorting machine design.

Expected From Students:

- Successfully build a basic circuit with an IR sensor, LED, resistor, and battery on a breadboard.
- Identify the correct terminals (polarity) of the IR sensor and the LED during circuit construction.
- Carefully reverse the terminals of the IR sensor and observe the resulting behavior of the LED (likely it will not light up).
- Understand that components in a circuit often have specific polarity requirements for proper function.
- Begin to think about the physical placement and connection of the IR sensor within their overall height sorting machine design as part of the "create" stage of the EDP.

Materials Needed:

- Student books with this "Practice" and "Showcase" page.

- IR sensors suitable for object detection (ensure they are compatible with a simple LED circuit).
- LEDs.
- Resistors (appropriate value for the LED and battery voltage).
- Batteries (with appropriate voltage for the LED).
- Breadboards and jumper wires.
- Circuit diagrams for the basic IR sensor and LED circuit.

Step-by-Step Instructions:

1. Practice - Building the IR Sensor Circuit (25-30 minutes):

- Introduce the "Practice" activity as a hands-on way to understand how to connect an IR sensor in a simple circuit.
- Provide clear circuit diagrams and step-by-step instructions for building the circuit with the IR sensor, LED, resistor, and battery on the breadboard.
- Circulate and provide assistance to students as they build their circuits, ensuring they are connecting components correctly and paying attention to polarity.
- Once the basic circuit is working (the LED lights up when the IR sensor detects an object or based on its resting state), instruct students to carefully reverse the terminals of the IR sensor.
- Ask them to observe and note what happens to the LED when the IR sensor's polarity is reversed (it should likely not function).
- Discuss why the correct orientation of components like sensors and LEDs is crucial in circuits.

2. Showcase - Integrating the IR Sensor (15-20 minutes):

- Transition to the "Showcase" section by referencing the robot's speech bubble and the EDP diagram.
- Remind students that they have now learned about IR sensors and how to connect them.
- Guide them to revisit their Engineering Design Process (EDP) documentation for their height sorting machine.
- Instruct them to focus on the "create" stage and brainstorm how they will physically add the IR sensor to their machine. This could involve considering:

- Where the sensor will be positioned to accurately measure the height of objects.
- How the sensor will be mounted or attached to the machine.
- How the sensor will be wired to the microcontroller (which they will likely work on in a future lesson).
- Encourage students to sketch or note down their initial ideas for integrating the IR sensor into their design.

Differentiation:

- **Support:** Provide students who struggle with circuit building with pre-wired sections or more simplified diagrams. Offer one-on-one assistance as needed.
- **Extension:** Challenge students who complete the basic circuit quickly to experiment with different resistor values and observe the effect on the LED's brightness or to investigate the sensor's detection range.

UNIT 02:

Chapter 01: Lesson 02 :Let's code

Time Frame: 90 MIN

Lesson Objectives:

- 1-Identify fundamental programming structures (variables, conditional statements, loops) essential for interacting with sensors.
- 2-Understand the basic syntax for reading analog input from an IR sensor using the Arduino IDE.
- 3-Write basic Arduino code to read raw data from an IR sensor and display it on the Serial Monitor.
- 4-Begin to implement conditional logic in Arduino code to interpret sensor data (e.g., classify height).
- 5-Relate the process of coding the sensor to the "create" stage of the Engineering Design Process (EDP).
- 6-Reflect on their ability to understand IR sensor principles and write basic Arduino code.

Challenge questions of the lesson :

- 1-What are the most important programming structures to consider when you want your Arduino to make decisions based on sensor input? Why?
- 2-How can you translate the raw data you get from a sensor into meaningful information that your program can use to control a machine?
- 3-Where does the coding process fit within the larger Engineering Design Process for building our height sorting machine?
- 4-What are some potential challenges you might encounter when trying to write code to accurately interpret sensor data and control a physical system?

here are short and simple answers to the challenge questions:

- Key code parts for decisions: IF statements help the Arduino decide what to do based on what the sensor says.
-
- Sensor data to machine actions: You need to tell the Arduino what the sensor numbers mean (like "high number = tall") so it can control the machine.
-
- Coding in the design process: Coding is part of building (creating) the machine's "brain."
-
- Coding challenges: It can be tricky to make the code understand the sensor correctly and make the machine do what you want.

SEL Relation:

- **Self-Awareness:** Students reflect on their initial understanding of programming and their comfort level with writing code.
- **Self-Management:** Students practice persistence and problem-solving skills when attempting to write and debug their code.
- **Social Awareness:** Students may collaborate with peers to understand coding concepts or troubleshoot issues.
- **Relationship Skills:** Students communicate their coding ideas and challenges with the teacher and classmates.
- **Responsible Decision-Making:** Students begin to understand how code dictates the behavior of a technological system and the importance of logical and accurate programming.

1-Let's think

Teacher Role:

- Introduce the shift from understanding sensors to programming them.
- Pose the "Let's Think" questions to initiate student thinking about the fundamental programming concepts needed to interact with a sensor.
- Set the stage for the "Explore" activity, where students will begin writing code to interpret IR sensor data and potentially control a virtual or physical output based on height.
- Direct students to the "Read about it" resource for guidance on coding with IR sensors and Arduino.

Expected From Students:

- Begin to consider the programming concepts (like variables, conditional statements, loops) that might be necessary to read and process data from an IR sensor.
- Think about how code instructions are executed sequentially to achieve a desired outcome (controlling actions based on sensor input).
- Understand the objective of the "Explore" activity: to write code that uses IR sensor readings to make decisions.
- Recognize the "Read about it" resource as a valuable tool for learning how to code with IR sensors and Arduino.

Materials Needed:

- Student books with this "Let's code" page.
- Access to computers with the Arduino IDE installed.
- (Ideally) Arduino boards and connected IR sensors (as used in the previous lesson) for immediate coding.
- Internet access for the "Read about it" article (either projected or individual access).

Step-by-Step Instructions:

1. Engage with "Let's Think" (10-15 minutes):
 - Start by reminding students of the IR sensor they explored in the previous lesson.
 - Pose the first question: "Now that we know how the IR sensor works, how do you think we can tell the Arduino to understand the information it's giving us? What are some basic things we need to tell a computer to do?" (Guide them towards ideas like storing values, making decisions based on values).
 - Introduce the term "programming structures."
 - Pose the second question: "Once we have these code components, how does the Arduino actually follow our instructions? In what order does it do things?" (Guide them towards the concept of sequential execution).

- Briefly discuss potential programming structures like variables to store sensor readings, if statements to make decisions based on height, and potentially for or while loops if they consider continuous monitoring.
2. Introduce "Explore - Do it" (5 minutes):
- Explain the "Explore" task: "Your goal now is to try and write Arduino code that reads the values from the IR sensor. Imagine the sensor is telling us the height of an object. Your code should then do something based on that height – for now, we can just display different messages on the Serial Monitor depending on whether the object is 'short,' 'medium,' or 'tall'."
 - Emphasize that this is an initial attempt and they will likely need to refer to resources.
3. Direct Students to "Read about it" (5 minutes):
- Point out the "Read about it" section.
 - Explain that this article provides specific information and examples of how to interface an IR sensor with an Arduino and write basic code to read its values.
 - Instruct them to open the article and use it as a guide for the "Explore" activity.
4. Initial Coding Attempt (20-30 minutes):
- Allow students time to start writing their Arduino code. Encourage them to:
 - Define a variable to store the IR sensor reading.
 - Use `analogRead()` (assuming an analog IR sensor) to get the sensor value.
 - Use `Serial.begin()` in `setup()` and `Serial.println()` in `loop()` to display values on the Serial Monitor.
 - (If they are ready) Introduce if or if-else statements to check the sensor value against arbitrary thresholds and print different messages (e.g., "Short object detected," "Tall object detected").

Arduino code:

```
// Define the analog pin connected to the IR sensor's output
const int irSensorPin = A0;    // Example analog pin for IR sensor
```

```
// Define thresholds for height categories (adjust these based on your sensor)
const int shortThreshold = 300;
const int mediumThreshold = 600;
```

```
void setup() {
```

```

Serial.begin(9600);
Serial.println("IR Sensor Height Classification:");
}

void loop() {
  // Read the analog value from the IR sensor
  int irValue = analogRead(irSensorPin);
  Serial.print("Raw Value (0-1023): ");
  Serial.println(irValue);

  // Classify the object height based on the thresholds
  if (irValue < shortThreshold) {
    Serial.println("Object: Short");
  } else if (irValue < mediumThreshold) {
    Serial.println("Object: Medium");
  } else {
    Serial.println("Object: Tall");
  }

  Serial.println("-----");
  delay(500); // Delay between readings
}

```

Differentiation:

- Support: Provide students who are struggling with very basic "skeleton code" that includes setup() for serial communication and a basic loop() that reads the analog pin and prints the value.
 - Extension: Challenge students who quickly grasp the basics to research how to calibrate their IR sensor to get more accurate distance readings and use those readings to define more precise height categories in their code.
-

2-Practice

Teacher Role:

- Encourage students to begin writing Arduino code to read their IR sensor and potentially classify object heights, building upon the foundational code examples provided or their own attempts.
- Guide students to think about the structure of their code and how it will fit

into the "create" stage of their Engineering Design Process (EDP) for the height sorting machine.

- Facilitate a discussion about the key programming structures they might use (variables, conditional statements, loops).
- Direct students to reflect on their learning in this lesson by reviewing the "Now I can..." statements.

Expected From Students:

- Begin writing Arduino code to read the analog or digital output from their IR sensor.
- Attempt to implement basic logic (e.g., using if statements) to interpret the sensor data, possibly classifying object heights.
- Consider how their code will be organized and integrated into the control system of their height sorting machine within the EDP framework.
- Reflect on their understanding of IR technology, its use in height measurement, their ability to write basic control code, and their problem-solving skills.
- Check the boxes next to the "Now I can..." statements they feel confident in.

Materials Needed:

- Student books with this "Assessment" page.
- Access to computers with the Arduino IDE installed.
- Arduino boards and connected IR sensors.
- (Optional) Example code snippets for reading IR sensors and basic height classification (as provided previously).
- Student Engineering Design Process (EDP) documentation.

Step-by-Step Instructions:

1. Practice - Initial Coding (25-30 minutes):
 - Introduce the "Practice" section as the time to start translating their

understanding of the IR sensor into actual Arduino code.

- Encourage students to use the basic code examples discussed or the "Read about it" article from the previous page as a starting point.
- Remind them of the task from the "Explore" section: to read the sensor and potentially display different messages on the Serial Monitor based on perceived height.
- Circulate and provide support as students write their code, helping them with syntax, logic, and troubleshooting.

2. Showcase - Planning Code Integration (15-20 minutes):

- Direct students to the robot's speech bubble and the EDP diagram.
- Explain that now that they are starting to code, they need to think about how this code will fit into the overall "create" stage of their height sorting machine.
- Prompt them to consider:
 - How their code will read the sensor data.
 - What programming structures (variables, if statements, loops) will be essential for making decisions based on height.
 - How the code will eventually control the sorting mechanism (even if they haven't learned that yet).
 - Where the code development fits within their EDP timeline.
- Encourage students to make notes or sketches in their EDP documentation about their initial code structure ideas.

3. Now I can... - Self-Reflection (10 minutes):

- Read each "Now I can..." statement aloud.
- Encourage students to reflect on their learning and check the boxes next to the statements they feel they can now do.
- Emphasize that this self-assessment helps them identify their strengths and areas where they might need further clarification or practice.
-

Differentiation:

- Support: Provide students who are struggling with more structured starter code or pseudocode outlines to guide their coding process.
- Extension: Challenge students who are coding quickly to explore more advanced concepts like calibrating their sensor in code or implementing more sophisticated height classification logic.

UNIT 03:

Chapter 01: Lesson 01 :Let's make a counter

Time Frame: 90 MIN

Lesson Objectives:

- 1-Understand the potential of AI and Data Science to enhance the height sorting machine project.
- 2-Identify basic programming structures in Python for creating simple programs.
- 3-Run and interact with a Python GUI created using tkinter.
- 4-Understand the role of a GUI in visualizing and potentially improving an engineered system.
- 5-Begin to connect the development of a user interface to the "improve" stage of the Engineering Design Process (EDP).
- 6-Reflect on their initial understanding of integrating machine learning into the sorting project.

Challenge questions of the lesson :

- 1-How might collecting and analyzing historical data from our height sorting machine help us make it "smarter"?
- 2-What are the benefits of having a graphical user interface (GUI) for a complex system like our sorting machine?
- 3-How does the process of creating a GUI relate to the Engineering Design Process, particularly the "improve" stage?
- 4-What are some initial ideas you have about how machine learning algorithms could optimize the sorting decisions of our machine?

here are short and simple answers to the challenge questions:

Smarter machine with data: Past data can help the machine learn to sort better over time.

GUI benefits: A GUI makes it easier to see and control the sorting machine.

GUI in design process: It helps us see problems and make the machine better ("improve").

ML for sorting: AI could help the machine learn the best way to sort different objects.

SEL Relation:

- **Self-Awareness:** Students reflect on their understanding of advanced concepts like AI and Data Science and their comfort level with Python GUI programming.
- **Self-Management:** Students practice exploring new programming concepts and tools (like tkinter) and managing their time to complete the tasks.
- **Social Awareness:** Students may discuss the potential impact of AI and automation on society and different industries.
- **Relationship Skills:** Students may collaborate with peers to understand Python code or troubleshoot GUI issues.
- **Responsible Decision-Making:** Students begin to consider how data-driven decisions (potentially through machine learning) can lead to more efficient and effective technological solutions.

Stepping Stone

Teacher Role:

- Highlight the advanced nature of this unit, emphasizing the integration of Python, Machine Learning, and Data Science into the height sorting machine project.
- Briefly review the necessary prior knowledge: basic Arduino and Python, the definition of Machine Learning, and sensor interfacing with data acquisition.
- Clearly state the learning outcome for this unit: exploring the use of machine learning to optimize sorting decisions using historical data.
- Pose the "Let's Think" question to encourage students to start considering how AI and Data Science can enhance their existing project.
- Introduce the "Explore" activity as a first step towards using Python for data interaction by creating and using a simple GUI counter that simulates IR sensor input. Provide the Python code below to the students. Guide them on how to run this code.

Expected From Students:

- Reflect on their existing knowledge of Arduino and Python programming.
- Recall the basic definition of Machine Learning and its potential applications.
- Remember their experiences with interfacing sensors and handling the data they produce.
- Begin to brainstorm ways in which AI and Data Science could improve the functionality and efficiency of their height sorting machine.
- Understand the "Explore" task as an introduction to using Python for data interaction through the creation and use of a simple GUI counter that simulates IR sensor input.
- Run the provided Python code and interact with the GUI counter.
- Understand how the button clicks simulate the IR sensor detecting objects and incrementing a count.

Materials Needed:

- Student books with this "Stepping Stone" page.

- Computers with Python and the tkinter library installed.
- A Python Integrated Development Environment (IDE) like IDLE, VS Code with Python extension, or Thonny.

Step-by-Step Instructions:

1. Review Prior Knowledge (10 minutes):

- Begin by briefly discussing the "You need to know..." section. Ask students to share what they remember about:
 - Basic Arduino programming (reading sensor data, controlling outputs).
 - Basic Python syntax and concepts (variables, functions, GUI basics if known).
 - The core idea of Machine Learning (algorithms learning from data to make predictions or decisions).
 - How they have previously connected sensors to microcontrollers and worked with the data generated.

2. Introduce Learning Outcomes (5 minutes):

- Clearly explain the "You will be able to..." statement, emphasizing that this unit will focus on using Machine Learning to make their height sorting machine smarter and more efficient by learning from past data.

3. Engage with "Let's Think" (10 minutes):

- Direct students to the "Let's Think" question: "How can we enhance our project and integrate AI and Data science with it?"
- Encourage a class discussion. Prompt their thinking with questions like:
 - "What kind of data might our height sorting machine collect as it sorts objects?" (e.g., height, time, success of sorting).

- "How could the machine learn from this past data?" (e.g., to adjust sorting thresholds, identify patterns).
- "What are some benefits of making our sorting machine 'smart' using AI?" (e.g., increased accuracy, faster sorting, adapting to different types of objects).

4. Introduce "Explore - Do it" (15-20 minutes):

- Explain the "Explore" task: "As a first step into using Python for data interaction, we are going to use a simple Python program that simulates
- an IR sensor counting objects. This will help you get more comfortable with Python GUI basics and how we can represent sensor input in software."
- Provide students with the following Python code:

```

5. import tkinter as tk
6.
7. class IRCounter(tk.Tk):
8.     def __init__(self):
9.         super().__init__()
10.
11.         self.title("IR Counter")
12.         self.geometry("300x200")
13.
14.         # Counter variable
15.         self.count = 0
16.
17.         # Label to display the count
18.         self.counter_label = tk.Label(self, text=f"Count: {self.count}",
font=("Helvetica", 24))
19.         self.counter_label.pack(pady=20)
20.
21.         # Button to simulate IR sensor detection
22.         self.detect_button = tk.Button(self, text="Simulate IR Detection",
command=self.increment_counter)
23.         self.detect_button.pack(pady=10)
24.
25.         # Reset button to reset the count
26.         self.reset_button = tk.Button(self, text="Reset", command=self.reset_counter)
27.         self.reset_button.pack(pady=10)
28.
29.         # Quit button
30.         self.quit_button = tk.Button(self, text="Quit", command=self.quit)
31.         self.quit_button.pack(pady=10)
32.
33.     def increment_counter(self):
34.         """Simulate the detection of an object and increment the count."""

```

```

35.         self.count += 1
36.         self.counter_label.config(text=f"Count: {self.count}")
37.
38.     def reset_counter(self):
39.         """Reset the counter to zero."""
40.         self.count = 0
41.         self.counter_label.config(text=f"Count: {self.count}")
42.
43. if __name__ == "__main__":
44.     app = IRCounter()
45.     app.mainloop()
46.

```

- Guide students on how to save this code as a .py file (e.g., ir_counter.py) and run it using a Python IDE or the command line (python ir_counter.py).
- Encourage them to interact with the GUI: clicking the "Simulate IR Detection" button to see the count increase and the "Reset" button to set it back to zero. Explain that each click of the "Simulate IR Detection" button represents the IR sensor detecting an object.

Differentiation:

- Support: Provide students who are new to Python with step-by-step instructions on how to run the code. Explain the basic elements of the Python code (importing the library, creating the window, buttons, and labels).
- Extension: Challenge students to modify the Python code (e.g., add a label to display the time of the last detection, change the button text, or explore other simple GUI elements). They could also think about how they might adapt this counter to store the data in a file.

Practice

Teacher Role:

- Guide students to begin creating a basic graphical user interface (GUI) using the provided Python code and the tkinter library. Explain that this GUI will eventually visualize the sorting process.
- Encourage students to run and interact with the provided GUI, understanding the function of the "Simulate Sort" button and the feedback text area.

- Connect the GUI development back to the Engineering Design Process (EDP), specifically the "improve" stage, suggesting that a GUI allows for better observation and refinement of the sorting mechanism in future lessons.
- Direct students to reflect on whether they have explored the overarching learning outcome for the unit related to machine learning.

Expected From Students:

- Run the provided Python code to create and interact with the basic Sorting GUI.
- Understand the function of the "Simulate Sort" button as a placeholder for the actual sorting simulation.
- Recognize the text area as a space where feedback on the sorting process will be displayed in later lessons.
- Begin to consider how a visual interface can help in the "improve" stage of the EDP by providing a way to observe and analyze the system's behavior.
- Reflect on their initial understanding of the potential of machine learning to optimize sorting decisions and check the "Now I can..." statement if they feel they have started this exploration.

Materials Needed:

- Student books with this "Assessment" page.
- Computers with Python and the tkinter library installed.
- Python Integrated Development Environment (IDE).
- The following Python code for the basic Sorting GUI:

Python

```
import tkinter as tk
```

```
class SortingGUI(tk.Tk):
```

```
    def __init__(self):
```

```
        super().__init__()
```

```

self.title("Sorting Simulation")
self.geometry("400x300")

self.process_label = tk.Label(self, text="Sorting Process...",
font=("Helvetica", 16))
self.process_label.pack(pady=20)

self.simulate_button = tk.Button(self, text="Simulate Sort",
command=self.simulate_sorting)
self.simulate_button.pack(pady=10)

self.feedback_text = tk.Text(self, height=5, width=30)
self.feedback_text.pack(pady=10)
self.feedback_text.config(state=tk.DISABLED) # Make it read-only

self.improve_button = tk.Button(self, text="Improve Algorithm",
command=self.open_improvement_window)
self.improve_button.pack(pady=10)

self.quit_button = tk.Button(self, text="Quit", command=self.quit)
self.quit_button.pack(pady=10)

def simulate_sorting(self):
    """Simulates the sorting process and displays feedback."""
    self.feedback_text.config(state=tk.NORMAL)
    self.feedback_text.delete("1.0", tk.END)
    self.feedback_text.insert(tk.END, "Simulating height
measurement...\n")
    self.feedback_text.insert(tk.END, "Analyzing data...\n")
    self.feedback_text.insert(tk.END, "Sorting objects...\n")
    self.feedback_text.config(state=tk.DISABLED)

```

```

def open_improvement_window(self):
    """Opens a new window to simulate algorithm improvement (for later
    lessons)."""
    improve_window = tk.Toplevel(self)
    improve_window.title("Improve Sorting Algorithm")
    improve_label = tk.Label(improve_window, text="This will be
    implemented in a later lesson.")
    improve_label.pack(padx=20, pady=20)

if __name__ == "__main__":
    app = SortingGUI()
    app.mainloop()

```

- Student Engineering Design Process (EDP) documentation.

Step-by-Step Instructions:

1. Practice - Running the Sorting GUI (30-35 minutes):
 - Introduce the "Practice" activity as the next step in their Python exploration – using a visual interface for their sorting project.
 - Provide students with the Python code for the "Sorting GUI."
 - Guide students on how to save this code as a .py file (e.g., sorting_gui.py) and run it using a Python IDE or the command line (python sorting_gui.py).
 - Encourage them to interact with the GUI: clicking the "Simulate Sort" button to see the text feedback and the "Improve Algorithm" button to see the placeholder window. Explain that this GUI will be further developed in later lessons.
2. Connecting to the EDP (10-15 minutes):
 - Refer to the robot's speech bubble and the EDP diagram.
 - Explain that this GUI is a tool that will be valuable for the "improve" stage of the EDP. Discuss how a visual interface will eventually help them:
 - Observe the sorting process and any data related to it.

- Identify potential issues or areas where the sorting algorithm could be better.
 - Allow for user interaction to test and refine the system.
 - Instruct students to make notes in their EDP documentation about their initial GUI and how it relates to the "improve" stage.
3. Now I can... - Self-Reflection (5 minutes):
- Read the "Now I can..." statement: "Explore using machine learning algorithms to optimize sorting decisions based on historical data."
 - Ask students to reflect on whether the discussions and activities in this lesson have started them thinking about how machine learning could be applied to their project. They should check the box if they feel they have begun this exploration, even if they haven't implemented any ML yet.

●

Differentiation:

- Support: Provide students who are new to Python GUI programming with clear instructions on how to run the code and point out the different parts of the interface (window, buttons, text area).
- Extension: Challenge students who run the basic GUI quickly to try and modify the text displayed when the "Simulate Sort" button is clicked or to explore changing the appearance of the GUI elements.

Analysis of the code :

IR Counter GUI -

This document provides a detailed that creates a graphical user counter using the Tkinter library. to simulate the detection of objects and increment a counter. Below is the full code with explanations for each function and line of code.

Code Analysis

analysis of a Python script interface (GUI) for an IR The GUI allows the user

Code Analysis

```
import tkinter as tk
```

Imports the Tkinter library, which is used for creating the GUI elements.

```
class IRCounter(tk.Tk):
```

Defines the IRCounter class, which inherits from tk.Tk, making it the main window of the GUI.

```
def __init__(self):
```

The constructor method that initializes the GUI window and its widgets.

```
    super().__init__()
```

Calls the constructor of the parent class (tk.Tk) to initialize the Tkinter window.

```
self.title("IR Counter")
```

Sets the title of the main window to 'IR Counter'.

```
self.geometry("300x200")
```

Sets the size of the main window to 300x200 pixels.

```
    self.count = 0
```

Initializes the counter variable to 0, which will keep track of the detections.

```
    self.counter_label = tk.Label(self, text=f"Count: {self.count}", font=("Helvetica", 24))
```

Creates a label widget to display the current count. The text is set using an f-string, and the font is set to Helvetica, size 24.

```
    self.counter_label.pack(pady=20)
```

Positions the label in the window with 20 pixels of padding above and below.

```
self.detect_button = tk.Button(self, text="Simulate IR Detection", command=self.increment_counter)
```

Creates a button labeled 'Simulate IR Detection'. When clicked, it triggers the increment_counter method.

```
    self.detect_button.pack(pady=10)
```

Positions the detect button in the window with 10 pixels of padding.

```
self.reset_button = tk.Button(self, text="Reset", command=self.reset_counter)
```

Creates a button labeled 'Reset'. When clicked, it triggers the reset_counter method.

```
self.reset_button.pack(pady=10)
```

Positions the reset button in the window with 10 pixels of padding.

```
self.quit_button = tk.Button(self, text="Quit", command=self.quit)
```

Creates a button labeled 'Quit'. When clicked, it closes the application by calling the quit method.

```
self.quit_button.pack(pady=10)
```

Positions the quit button in the window with 10 pixels of padding.

```
def increment_counter(self):
```

Defines the increment_counter method, which increments the count variable and updates the label.

```
    self.count += 1
```

Increments the count variable by 1.

```
    self.counter_label.config(text=f"Count: {self.count}")
```

Updates the label text to display the new count.

```
def reset_counter(self):
```

Defines the reset_counter method, which resets the count variable and updates the label.

```
    self.count = 0
```

Resets the count variable to 0.

```
    self.counter_label.config(text=f"Count: {self.count}")
```

Updates the label text to display 'Count: 0'.

```
if __name__ == "__main__":
```

Checks if the script is being run directly (not imported as a module).

```
    app = IRCounter()
```

Creates an instance of the IRCounter class, initializing the GUI.

```
    app.mainloop()
```



Starts the Tkinter event loop, which keeps the application running and responsive to user actions.

Pyramakerz®

Height sorting project

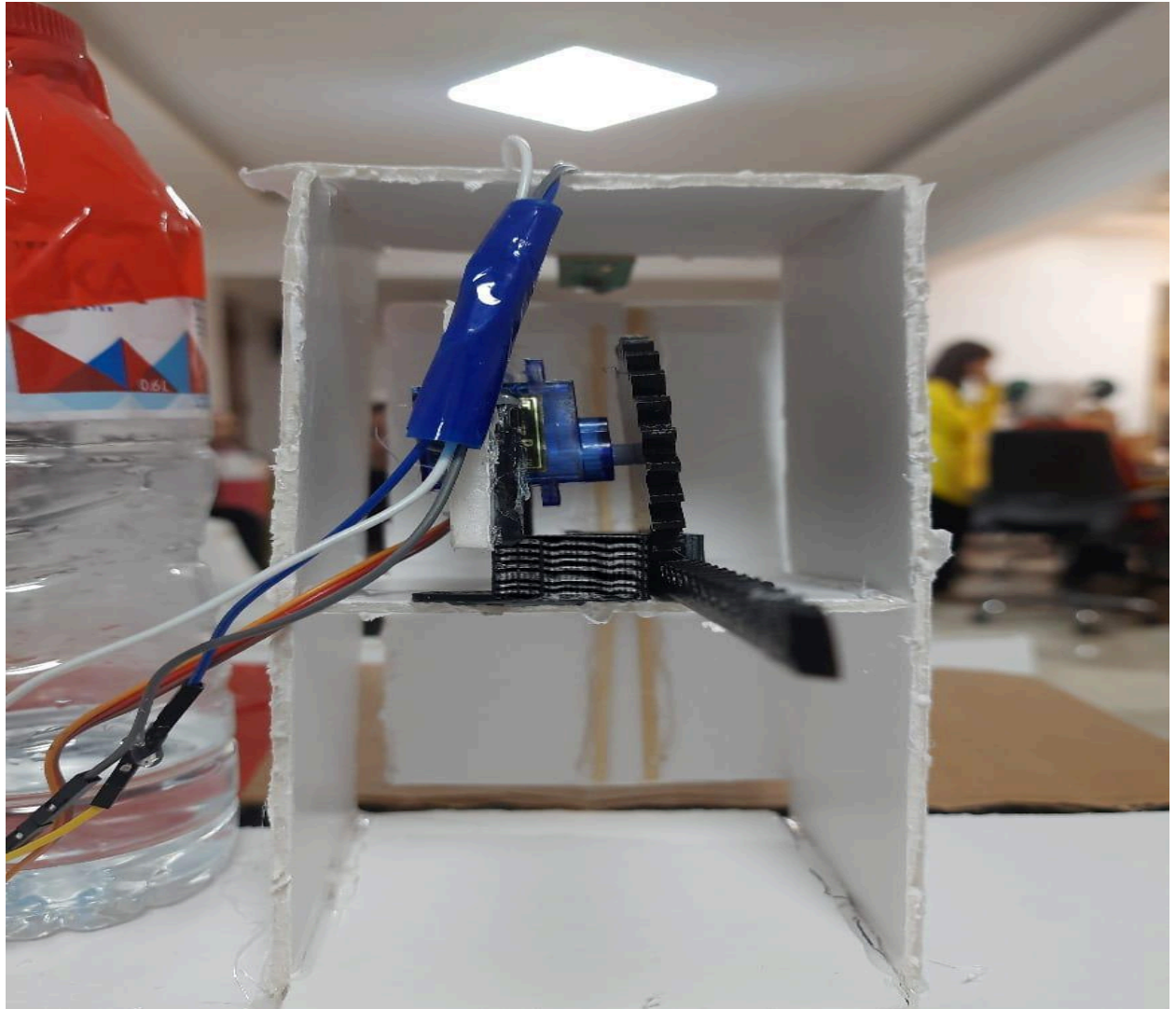
Grade 9:

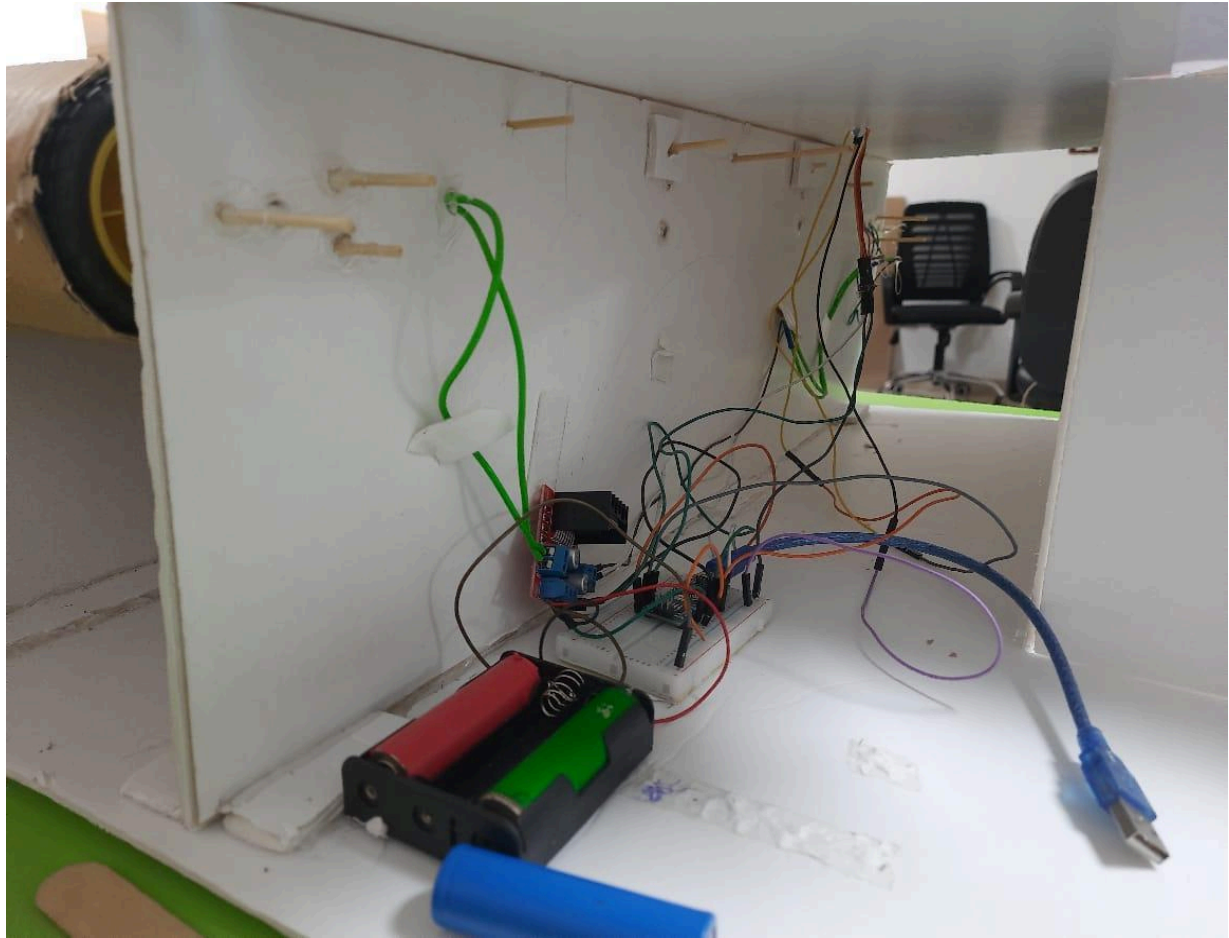
Video:

<https://www.instagram.com/reel/C6JgsUcLhWM/?igsh=cDJ5ZnFqcTMxaWNw>









Project idea:

The height sorting project involves students designing and building a device that can automatically sort a group of objects (such as plastic cups or blocks) by their height , design a mechanical system to physically sort the objects, and build a prototype to demonstrate their solution

Learning Objective:

- Apply problem-solving skills to design a mechanical system that can sort objects by height.
- Gain experience in prototyping and constructing a functional physical device using materials like cardboard, plastic tubes, 3D printed parts, etc.
- Improve teamwork and communication skills through collaborative design and construction of the project.
- Practice technical writing and presentation skills to effectively explain the design process and demonstrate the working prototype.

Components:

1. IR sensor
2. Arduino
3. Servo motor
4. 4 Dc motors
5. Jumpers
6. Motor drive
7. 4 wheels
8. Breadboard

Materials:

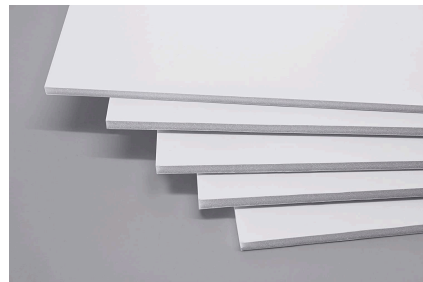
White cardboard



Brown cardboard



Foam board



plastic tubes



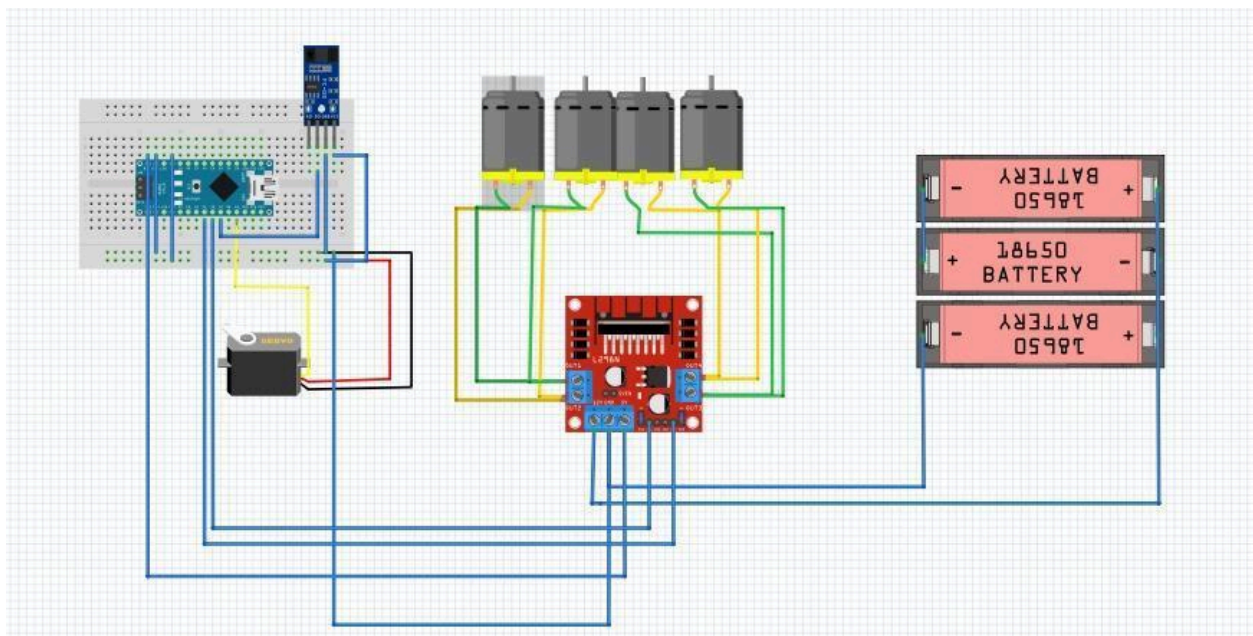
glue sticks



BBQ sticks

Wires

Circuits used:



Code:

```
#include <Servo.h>

int IRSensor = 7;
int pos = 0 ;
int motor1 = 5;
int motor2 = 6;
int sensorStatus =0;

Servo myservo;

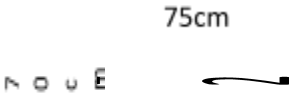
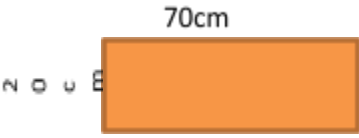
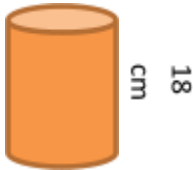
void setup() {

  Serial.begin(9600);
  myservo.attach(9);
  pinMode(IRSensor, INPUT);
  pinMode(motor1, OUTPUT);
  pinMode(motor2, OUTPUT);

}
void motor_stop()
{
  analogWrite(motor1,0);
  analogWrite(motor2,0);
}
void motor_start()
{
  analogWrite(motor1,120);
  analogWrite(motor2,120);
}
```

```
void loop() {  
  
    Serial.println(IRSensor);  
    sensorStatus = digitalRead(IRSensor);  
  
    if (sensorStatus == 1) // Check if the pin high or not  
    {  
        pos = 180;  
        myservo.write(pos);  
        delay(30); // print Motion Detected! on the serial monitor window  
        motor_stop();  
    }  
    else {  
        pos = 0;  
        myservo.write(pos);  
        delay(30);  
        motor_start();  
    }  
}
```

For the design:

Steps	shape
For the base of the design we used white hard cardboard	
For the 2 sides of the belt we used white hard	
For the belt we used the brown cardboard we peeled the outer layer of it to make teeth	
For the rollers to move the belt we made 5 cylinder by tubes with length = 18 cm	
For the connection between wheels cut 2 rectangular parts with dimensions W=12 , L=17 cm then we rolled it	

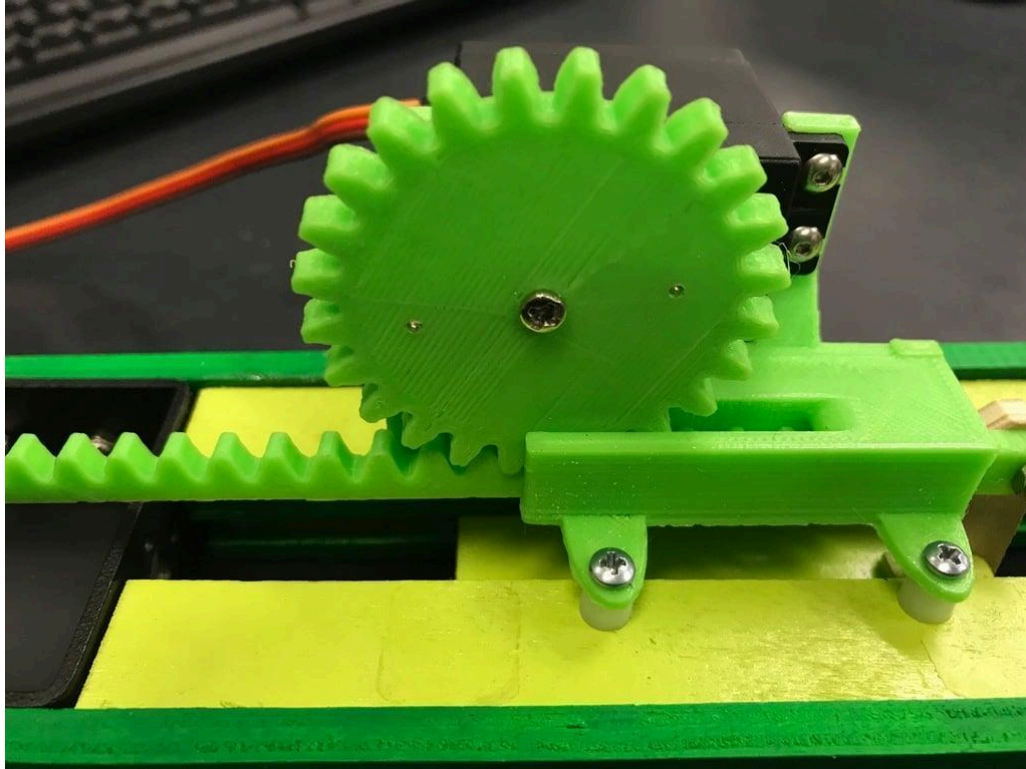
For the connection between two wheels we made this cylinder



Watch this:

<https://youtu.be/8Vnoso8rhaw?si=e8uxPd1ifZGblbZI>

For the pushing part we used rack and pinion gears with servo motor.



You can download the design from here:

<https://www.thingiverse.com/thing:3653846>